

Bare Metal System Backup and Recovery Using Open Source Tools

Disaster often strikes when least expected. When it's a question of safeguarding data against disk crash or corruption, it's best to have a backup and recovery system. Bare metal system recovery is particularly advantageous in that there is no need to even load the operating system to effect data recovery.

Backup and recovery of heterogeneous data files is one of the key aspects of personal as well as corporate computing. In our routine tasks, we create and update a number of files in our systems. But the major problem occurs when the operating system crashes or any media stalls. This issue generates huge complications and, many a time, our work stops. Often, we have to create the files using backup, recovery and disaster management tools and techniques.

A number of tools are available in the open source as well as the proprietary domain, which provide secured backup, fast recovery and fault tolerance.

The key challenge in disaster management arises when the operating system crashes and when we are not able to load the OS to execute the recovery tool. In such scenarios, we can work with bare metal backup and recovery tools so that there is no need to even load the operating system. Using these tools, the backup, recovery and disaster management can be achieved at the BIOS level itself.

Features of effective backup and fault-tolerant tools

Some of the major features generally required for personal as well as corporate computing are:

- Bare metal recovery and fault tolerance
- Non-redundant or de-duplicated backup
- Security and privacy using multi-level encryption
- Dynamic compression and archiving
- Dynamic repository management
- Automatic/scheduled backups
- Cloud based backup and recovery
- Import and export for multiple platforms
- Version tracking

- Configurability with scope for extension using plugins
- Command line as well as GUI panels for usability
- Transaction fault tolerance to avoid data loss



- Being volume oriented to support compression, splitting and merging for multiple devices and platforms
- Malware scanning
- Universal view and updates
- Cross platform
- Support for multiple data formats
- Support for multiple databases
- Generation of reports, alerts and logs

Free and proprietary software backup tools

A number of software tools are being used in personal as well as commercial domains for backup of files and documents. Table 1 shows Linux based free software with command line interfaces.

Program/Tool	Base language
AMANDA	C and Perl
Areca Backup	Java
duplicity	Python
rdiff-backup	Python
Attic	Python
Bacula	C++
Back In Time	Python
DAR	C++
luckyBackup	C++
Mondo Rescue	C and UNIX Shell
obnam	Python
Synching	Go
Unison	OCaml

Table 1